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Paper Code : PEC-CS85- S01 B/PEC-IT501B Artificial Intelligence
UPID : 005510

CS/B.TECH(N)/DDD/SEM-5/5510/2025-2026/1024

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

[1 x 10 = 10]

1. Answer any ten of the following :

- ☒ (i) What is Intelligence?
- ☒ (ii) What do you mean by simulated annealing in AI?
- ☒ (iii) Define knowledge base.
- ☒ (iv) What do you understand by FOL or First Order Logic?
- ☒ (v) Define Syntax.
- ☒ (vi) What is an agent in AI?
- ☒ (vii) What is Plateau/flat local maximum in the State Space Diagram?
- ☒ (viii) What do you understand by Shoulder in the State Space Diagram?
- ☒ (ix) "Mary loves everyone." [assuming domain (D) includes both humans and non-humans] - Convert into 1st-order predicate logic.
- ☒ (x) What is Relevance based Learning (RBL) in AI?
- ☒ (xi) Define Artificial Intelligence in terms of "Thinking Humanly".
- ☒ (xii) What do you mean by A* search?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

- ☒ 1. What is depth limited search? Explain with an example. [5]
- ☒ 2. Explain Mini-Max algorithm in Artificial Intelligence. [5]
- ☒ 3. Design the PEAS information for taxi satellite image analysis system agent and part-picking robot agent. [5]
- ☒ 4. What is Alpha Beta Pruning? What are the condition for Alpha Beta Pruning? What is Adversarial Search? [5]
- ☒ 5. What is the main purpose of expert systems? Describe the phases of designing an expert system? What term is used to call the whole process? [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

- ☒ 7. (a) What is knowledge representation? [5]
(b) Represent various types of knowledge in AI. [10]
- ☒ 8. Write Short notes on [15]
(a) Propositional logic in Artificial Intelligence
(b) First-Order Logic in Artificial Intelligence
(c) Difference between Inductive and Deductive reasoning
- ☒ 9. (a) What is NLP or Natural Language Processing? [2]
(b) What are the pros and cons of NLP? [5]
(c) What are different phases of NLP? [5]
(d) Write down the components of NLP. [3]
- ☒ 10. (a) Draw the state space graph of Hill climbing search. What are the draw backs of this algorithm? [11]
(b) How to Over Come the above drawbacks? [4]
- ☒ 11. (a) Explain Goal based Agent, Utility based Agent, and Learning Agent architecture with proper diagram. [10]
(b) Explain Learning Agent architecture with proper diagram. [5]

*** END OF PAPER ***